GG-W-001253(DSA-DM)
1 May 1967
Interim Revision of
Fed. Spec. GG-S-809
21 October 1947

INTERIM FEDERAL SPECIFICATION

WIRE, STEEL, CORROSION RESISTING (SURGICAL SUTURE AND DENTAL LIGATURE)

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification GG-S-809.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers corrosion-resisting steel wire for use in suturing and ligating in surgical procedures.

1.2 Classification. Wire shall be of the following types and sizes as specified (see 6.2):

Type I. - Surgical suture

Size 1. - 0.009 inch in diameter.
Size 2. - 0.014 inch in diameter.
Size 3. - 0.018 inch in diameter.
Size 4. - 0.028 inch in diameter.

Type II. - Dental ligature

Size 5. - 0.020 inch in diameter.

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2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

STANDARDS

Federal

FED-STD-66	Steel: Chemical Composition and Hardenability.
FED-STD-123	Marking for Domestic Shipment (Civilian Agencies).
FED-STD-151	Metals; Test Methods.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

(Copies of Military specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. Wire shall be fabricated of AISI 304 stainless steel in accordance with FED-STD-66.

3.2 Style, design, and dimensions.

3.2.1 Type I. Sizes 1, 2, and 3 wire shall measure 0.009 inch, 0.014 inch, and 0.018 inch in diameter, respectively. For sizes 1, 2, and 3 wire, one continuous length of wire, weighing not less than one ounce, shall be neatly wound on a wood or plastic spool of adequate size with both ends of the wire securely attached to the spool. Size 4 wire shall measure 0.028 inch in diameter. One continuous length of size 4 wire weighing not less than 4 ounces shall be wound neatly in a coil.

3.2.2 Type II. Size 5 wire shall measure 0.020 inch in diameter and shall be furnished in packages of fifty (50) 18-inch (minimum) strands.

3.2.3 Diameter tolerance. A minus tolerance of 0.0005 inches and no plus tolerance shall be allowed in the diameters of all types and sizes of the wire.

3.3 Tensile strength. All types and sizes of the wire shall have a minimum tensile strength of 80,000 pounds per square inch (p.s.i.) and a maximum tensile strength of 120,000 p.s.i. when tested as specified in 4.4.1.

3.4 Elongation. All types and sizes of wire shall have a minimum elongation of 35 percent in a 10-inch gage length when tested as specified in 4.4.2.

3.5 Corrosion resistance. Wire shall show no corrosion when subjected to the boil test as specified in 4.4.3.

3.6 Finish. Wire shall be furnished in the bright annealed or dead soft temper, and shall have a bright, smooth finish.

3.7 Passivation. The wire shall receive a passivation treatment after final processing.

3.8 Identification marking. Marking for Type I wire shall include size number, diameter, and weight of wire as applicable for the specific size. For Type I, sizes 1, 2, and 3 wire, this information shall be marked on both sides of each spool. For Type I, size 4 wire, the information shall be marked on tags and each coil tagged in at least two places.

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3.9 Workmanship. Workmanship shall be first class throughout. The wire shall be of uniform quality and temper, smooth, clean, bright, and free from kinks, waviness, seams, tarnish, or other defects which may detract from the appearance or impair the serviceability of the wire.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The acceptable quality levels shall be as indicated in table I.

Table I. Inspection and Acceptable Quality Levels

Type defect	Inspection level	AQL (percent defective)
Major	II	1.0
Minor A	II	2.5
Minor B	S-2	4.0

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105. The unit of product shall be one spool (for Type I, sizes 1, 2, and 3), one coil (for Type I, size 4), or one package (for Type II, size 5). A minimum of two (2) determinations shall be made on each lot tested, and the length of the specimen tested shall be 10 inches.

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Table II. Tests

Characteristic	Requirement	Test procedure	Inspection level	AQL (percent defective)
Tensile strength	3.3	4.4.1	S-1	*
Elongation	3.4	4.4.2	S-1	*
Boil test	3.5	4.4.3	S-1	*

* Acceptance number shall be zero for all sample sizes.

4.2.3 Sampling for raw materials. Sufficient quantity of each lot of metal used in the construction of the wire shall be selected for chemical analysis.

4.2.4 Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

4.3 Examination. The wire shall be examined to determine compliance with all requirements contained in the specification. Examination shall be conducted as specified in 4.2.1. Any wire in the sample containing one or more defects shall be rejected, and if the number of defective wires in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Inspection shall not be restricted to the defects in table III.

Table III. Classification of Defects

Categories	Defects
Critical:	None defined
Major:	
101	Type and size (diameter) not as specified.
102	Wire not of uniform quality or finish.
103	Workmanship defects present such as kinks, waviness, seams, or tarnish.
104	Finish not as specified.
Minor A:	
201	Identification marking not as specified (Type I only).
202	Packaging and marking not as specified.
Minor B:	
251	Weight (Type I) or length (Type II) not as specified.
252	Number of strands in unit package (Type II only) not as specified.

4.4 Tests.

4.4.1 Tensile strength. Tests for tensile strength shall be conducted on samples in "as received" condition in accordance with method 211 of FED-STD-151. The testing machine shall be operated at a uniform rate of speed which shall move the heads or jaws apart at the rate of one inch per minute.

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4.4.2 Elongation. The elongation shall be determined on the samples used for the test for tensile strength. A ten inch gage shall be used. Samples which break outside the gage length may be discarded. The determined value shall be recorded to the nearest 0.5 percent.

4.4.3 Soil test. Wire shall be tested by the following method. Before testing, wire shall be scrubbed with soap and water, rinsed thoroughly in distilled water, and dried. After cleaning, the wire shall be boiled in distilled water in a glass beaker for 30 minutes, and allowed to remain submerged for an additional 24 hours.

4.4.4 Chemical analysis. Chemical analysis shall be conducted in accordance with Methods 111 or 112 of FED-STD-151.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One spool (1 ounce of wire), one coil (4 ounces of wire) or one package (50 wires), as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Size	Unit package	Intermediate package	Exterior container
1	1 unit	12 units	288 units
2	1 unit	12 units	288 units
3	1 unit	12 units	288 units
4	1 unit	12 units	96 units
5	1 unit	12 units	60 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Unit package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

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5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable, or PPP-B-636. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Civilian agencies. Marking of shipments shall be in accordance with FED-STD-123.

5.3.2 Military agencies.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. In addition, the unit package for size 5 wire shall include the size number and diameter of the wire.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Wire shall be suitable for the intended use specified below:

Type I, Size 1. - Suitable for use in ligation of vessels.

Type I, Sizes 2, 3, and 4. - Suitable for use as a suture material for surgical procedures.

Type II, Size 5. - Suitable for anchoring orthodontic appliances in the oral cavity.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Type and size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available,but is intended to cover the types which are normally procured to meet requirements.

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6.4 Supercession data. The wire covered by this Interim Federal Specification corresponds to the wire covered by GG-S-809, as shown in table V.

Table V. Corresponding Classification of Wire

GG-S-809		GG-W-001253(DSA-DM)	
Class	Size	Type	Size
A	1	I	1
A	2	I	2
A	3	I	3
B	4	II	5
B	5	I	4

Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

Fold

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION: QQ-W-001253(DSA-DM)

ORGANIZATION:

CITY AND STATE:

CONTRACT NUMBER:

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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